Nicholas Kamra

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Github | Personal Website | LinkedIn | Blog

EDUCATION

Concordia University | Montreal, Quebec
Bachelor of Science in Computer Science, September 2022 - December 2025
GPA: **3.56 (Distinction Award)**

PROJECTS

Flappy Bird Clone with Genetic Neural Network (C# and Unity)

- Developed a Flappy Bird clone using Unity, incorporating a custom Genetic Neural Network from scratch; Achieving scores above human ability - theoretically able to score to **infinity**.
- Designed the network with 1 hidden layer (6 neurons, ReLU activation) and 1 output neuron (Sigmoid function) to optimize agent performance.

Digit Recognizer with Multi-Class Classification (Unity, Pytorch, NumPy, Matplotlib, pandas)

- Designed and trained a PyTorch model for multi-class classification on the MNIST dataset, achieving **98%** accuracy; Implemented data augmentation to handle noise.
- Integrated the model into Unity using my **.NET, C# Supervised Learning Neural Network Library** to store weights and biases and handle forward passes for predictions.
- Created a custom canvas in Unity for users to draw digits and use the model to make predictions.

MLOps Animal-10 ResNet32 Classifier (PyTorch, FastAPI, Docker, GCP)

- The PyTorch ResNet32 architecture was pre-trained on the ImageNet 1k dataset and fine tuned on the Animal-10 dataset on Kaggle, in which EDA was used to explore the imbalances of the classes and example images were randomly selected from the dataset and observed. The model achieved a cross-validation accuracy of **90.06%** and an F-1 score of **0.9**. Accuracies and loss per epoch were graphed using Matplotlib.
- Created a full MLOps pipeline, including CI/CD with Github Actions, deploying Docker images on Google Cloud Platform in the Artifact Registry, where the model predictions FastAPI methods were exposed on Google Cloud Run.

EXPERIENCE

Lead Software Engineer | PETCH | Portland, ME: Feb 2026 - May 2026

- Built the initial software from the ground up for SMS campaigns for pre-visit and after-visit veterinarian communication. Utilized AWS services such as App Runner, Step Functions, and Elastic Container Registry in order to host apps and campaigns, and Docker, FastAPI, and Github Actions to enable CI/CD and containerization of our application APIs; Github was used for version control.
- Lead a small team of two other engineers, distributing tasks, and crafting the entire architecture.
- Achieved a customer satisfaction rate of **95%**, and the product saved vets an average of **3 hours** per week by shifting their manual follow-ups onto our automated campaigns.

AI Apprentice | AI Launch Lab Mentorship Program | Montreal, QC: Oct 2024 - May 2025

- Worked with mentors within the AI field and my cohort to learn the many different aspects of machine learning. This includes but is not limited to: neural network architectures; bias-variance trade-off; regression models; activation functions; basics of LLMs. Weekly quizzes put us against each other to gauge our understanding. With our cohort teams, we built small projects which we presented to our mentors at the end of the season.

SKILLS

- **Languages**: Python, C#, Java, SQL, C++, JavaScript, HTML5, CSS
- **Frameworks/Libraries/Engines**: PyTorch, pandas, Matplotlib, scikit-learn, Jupyter Notebooks, NumPy, Google Cloud Platform (GCP), Docker, Unity, AWS (Step Functions, ECR, App Runner)